INSIDE THE EARTH

Interpreting earthquake patterns through geography, time, depth and magnitude

DSC5002 Project Team | City University of Hong Kong | USGS catalogue, 2000–2025

ABSTRACT We examine 46,463 M ≥ 5 earthquakes using geographic reference points, equal-duration time comparisons and local depth sections. Events concentrate near mapped plate boundaries, but period differences also coincide with changes in catalogue composition. Regional depth tails and selected event sequences reveal structure that pooled summaries conceal. Our results describe a fixed catalogue; they do not estimate future hazard.

01 STUDY DESIGN

Why combine these views?

Earthquakes vary in location, depth, magnitude and timing. A map identifies clusters, yet it cannot explain whether they reflect geography, a few large sequences or changes in reporting. We connect these views to ask where recorded activity concentrates, how it varies over time, and which regional structures survive beyond global averages.

Table 1. Common analysis design

Element	Definition
Window	2000–2025 UTC
Selection	Earthquake; preferred M ≥ 5
Observations	46,463 unique event IDs
Regions	6 ordered boxes + Other
Distance	PB2002 spherical arcs
Depth classes	Shallow <70; intermediate 70–<300; deep ≥300 km

Processing and data quality

Coordinates, magnitude and depth are present for every record. Four negative reported depths are retained. Event IDs are deduplicated and non-earthquake types are excluded. A complete download does not prove complete earthquake detection, nor does a preferred magnitude represent one homogeneous scale.

The spatial setting

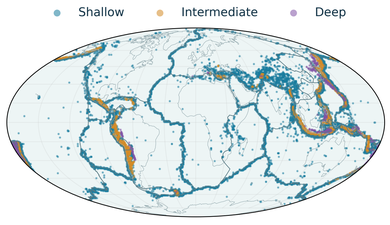


Fig. 1. All epicentres on an equal-area Mollweide map; colours denote depth classes and grey lines PB2002 boundaries. Overlapping points are not area-normalized density.

Analytical strategy

Use one frozen sample, explicit denominators and sensitivity checks. Compare recorded patterns while keeping geological interpretation and measurement limitations separate.

02 SPACE & REFERENCE

A reference for concentration

Boundary proximity needs a geographic denominator. We compare earthquake distances with points uniformly distributed over Earth's surface, including oceans. A second reference keeps the observed latitude distribution and varies longitude.

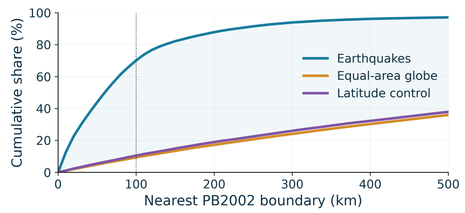


Fig. 2. Cumulative distance distributions use the same PB2002 arcs and spherical geometry (R = 6,371.0088 km) for events and both references.

Table 2. Within-distance shares (%)

Radius (km)	Events	Equal area	Latitude control
50	43.92	4.96	5.54
100	70.15	9.42	10.50
200	87.90	17.27	18.99
500	97.13	35.98	37.88

How to interpret the contrast

70.15% of events lie within 100 km, compared with **9.42%** of equal-area points and **10.50%** of latitude-controlled points. The observed-to-area share ratio is **7.45**. Thus, concentration remains clear under two geographic references; the ratio is not a risk multiplier or a causal effect.

Numerical and geological limits

Four independent 16,384-point designs give 9.38–9.53% within 100 km. This is numerical sensitivity, not a confidence interval. PB2002 is a simplified boundary layer, not a complete map of active faults.

03 TIME & COMPARABILITY

Time and threshold both matter

Annual M ≥ 5 counts fluctuate substantially, peaking at **2,701 in 2011**. Comparing equal 13-year periods gives a different summary from fitting a trend across the whole record.

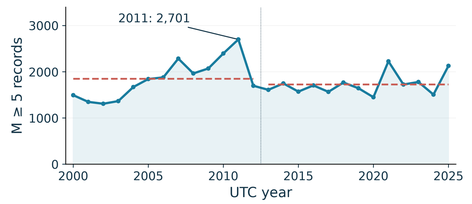


Fig. 3. Annual counts and period means (dashed). Rate comparisons allow for 4,749 days in 2000–2012 and 4,748 days in 2013–2025.

Table 3. Catalogue comparability

Indicator	2000–2012	2013–2025
M ≥ 5 records	24,030	22,433
Annual mean	1848.5	1725.6
Mw-family share	73.45%	55.05%
Horizontal-error availability	0.28%	86.80%

A change in records is not enough

The later-period record rate is **6.63% lower at M ≥ 5** and **15.42% lower at M ≥ 6**. Yet the full-period M5 Sen slope is positive. Magnitude-type composition and error-field availability also change, so these contrasts alone cannot establish a change in physical seismic activity.

Regional and window sensitivity

M5 rate changes differ: Indonesia/Philippines −25.58%; Mediterranean/Himalaya +8.82%. Removing 30-day / 500-km windows around 28 M ≥ 8 events changes the global contrast to −3.37%. This is window sensitivity, not declustering.

04 DEPTH & ASSOCIATIONS

Similar medians, different tails

Globally, **83.20%** are shallow. Japan/Kuril and Tonga/Fiji share a 33 km median, but their deep-event shares differ by more than twofold.

Table 4. Regional depth summaries

Region	N	Median (km)	Deep (%)
Global	46,463	23.7	3.71
Japan / Kuril	4,954	33.0	5.01
Tonga / Fiji	8,904	33.0	11.76
Andes	3,769	35.0	2.60

Local sections reveal structure

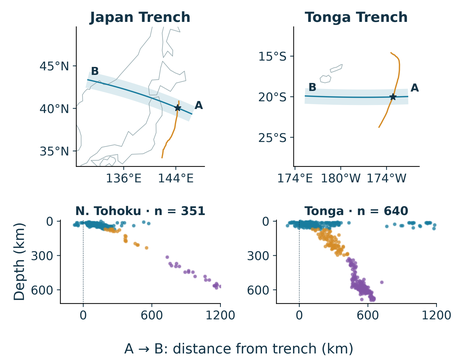


Fig. 4. Maps: gold trench, blue route/swath, star at 0 km. A → B spans −200 to +1,200 km; swath ±100 km. Depth and distance use equal kilometre scales. Routes were selected after exploration.

Associations and interpretation

Selected-section deep counts are **21/351** in northern Tohoku and **186/640** in central Tonga. Across the full regions, depth–boundary-distance Spearman ρ is **+0.39 globally** and **+0.56 in Japan/Kuril**. Magnitude associations are weak overall. These relationships describe patterns; pooled coefficients and illustrative sections do not identify a causal mechanism.

05 MAGNITUDE & SEQUENCES

Rare events, large proxy share

Event frequency and a magnitude-based energy proxy answer different questions. Their shares are compared directly below.

Table 5. Counts and relative proxy

M band	N	Events (%)	Proxy (%)
5–<6	42,589	91.66	2.00
6–<7	3,487	7.50	5.52
7–<8	359	0.77	21.27
≥8	28	0.06	71.21

The **28 events at M ≥ 8** contribute **71.21%** of the summed proxy, $R(M)=10^{1.5(M-5)}$. One magnitude unit multiplies R by 31.6. Mixed preferred scales mean that this is illustrative, not measured joules, damage or risk.

Table 6. Fixed-window case counts

Case	Before	After	First 3d (% after)
Sumatra 2004	2	143	60.1
Maule 2010	1	302	60.9
Tohoku 2011	33	563	70.9

Radius ≤500 km; UTC windows [−30,0) and [0,30) days from the exact anchor time. Anchors are excluded; all windows are fully observed.

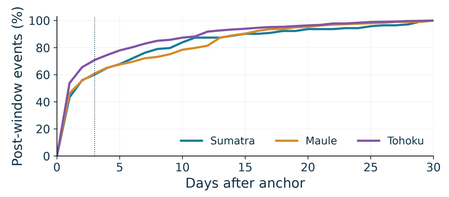


Fig. 5. Each curve uses its own 30-day post-window total. Day d counts events in [0,d), separating timing from total counts.

Scope: These selections do not prove causal aftershock membership or a universal sequence law.

CONCLUSION Geography, regional depth tails and clustered sequences give a coherent descriptive picture. Time contrasts require catalogue-comparability checks; reference choices and window definitions remain central to interpretation. The strongest result is a transparent comparison of recorded patterns, not a forecast of future hazard.

[1] USGS FDSN · [2] Bird (2003), PB2002; GeoJSON (ODC-By) · [3] Natural Earth 110m (public domain) · [4] USGS magnitude–energy guidance. Reproducible methods, all figures and sensitivity results: project repository.